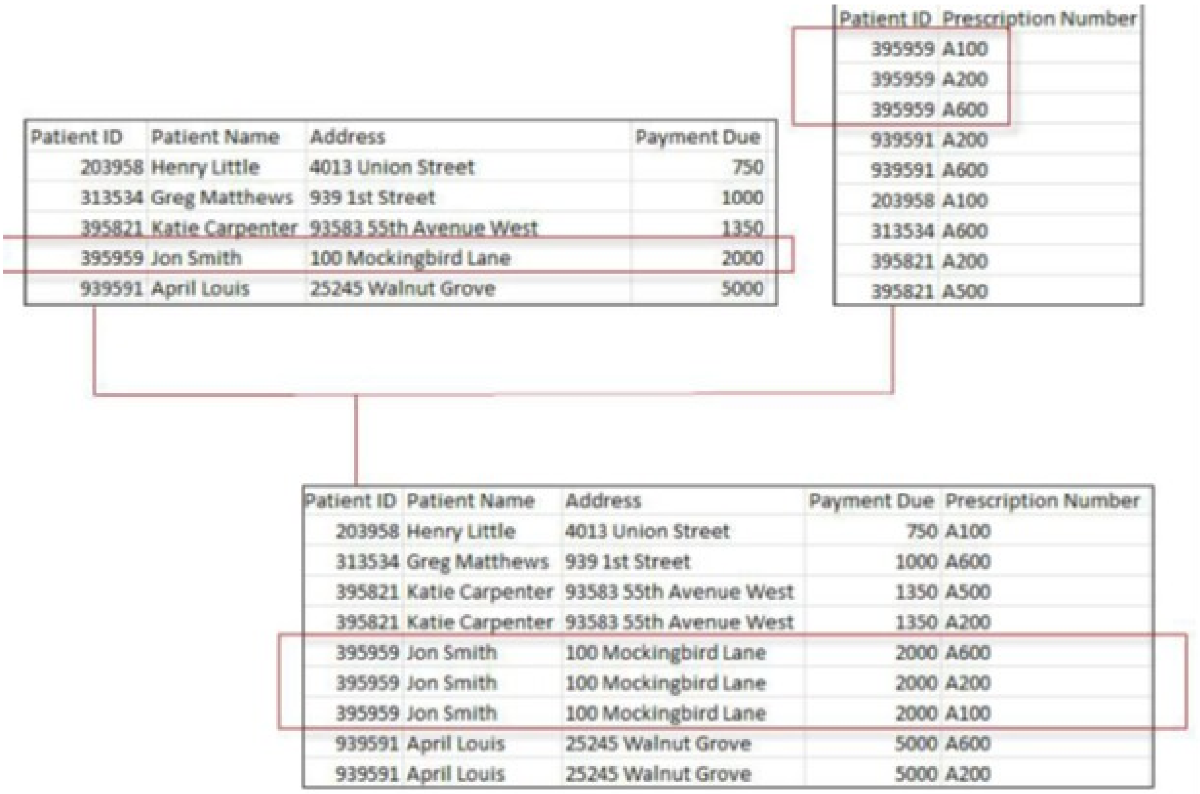
Duplication of Records in Tableau

Cardinality may affect the results returned when joining tables because joining tables that have **one-to-many** or **many-to-many** relationships can have the effect of duplicating values or multiplying rows by the number of times a primary key’s match appears in the secondary table.

In the diagram below you can see the left table has a one-to-many relationship with the right table because there are multiple prescriptions for the same patient ID. When these tables are joined, the result table shows that for every time Patient ID 395959 is matched with a prescription in the right table, the values from the left table duplicated the same number of times. Although the payment due for Patient 395959 should be $2000, aggregating the values in the result table produces the incorrect amount of $6000 $2000 x 3 or the number of times the patient ID appears in the right table).



Patient ID	Patient Name	Address	Payment Due
203958	Henry Little	4013 Union Street	750
313534	Greg Matthews	939 1st Street	1000
395821	Katie Carpenter	93583 55th Avenue West	1350
395959	Jon Smith	100 Mockingbird Lane	2000
939591	April Louis	25245 Walnut Grove	5000

Patient ID	Prescription Number
395959	A100
395959	A200
395959	A600
939591	A200
939591	A600
203958	A100
313534	A600
395821	A200
395821	A500

Patient ID	Patient Name	Address	Payment Due	Prescription Number
203958	Henry Little	4013 Union Street	750	A100
313534	Greg Matthews	939 1st Street	1000	A600
395821	Katie Carpenter	93583 55th Avenue West	1350	A500
395821	Katie Carpenter	93583 55th Avenue West	1350	A200
395959	Jon Smith	100 Mockingbird Lane	2000	A600
395959	Jon Smith	100 Mockingbird Lane	2000	A200
395959	Jon Smith	100 Mockingbird Lane	2000	A100
939591	April Louis	25245 Walnut Grove	5000	A600
939591	April Louis	25245 Walnut Grove	5000	A200

Solution:-

Tableau offers several solutions for addressing duplication of records when joining tables that have a one-to-many or many-to-many relationship.

1) Changing the aggregation from SUM to AVG, MIN, or COUNTD may sometimes resolve the issue.
2) Other times, creating a calculated field, using a table calculation or choosing to blend the data will provide a solution. 3) Custom SQL is another way to remove duplicates.

The technique you use is dependent upon the data as well as what you are trying to accomplish.

Nested Table Calculations behaviour

A nested table calc is when one table calc is used inside another table calc, which is different from writing a calculation that uses two table calculation functions independently. This is an important distinction because a nested table calc will allow you to compute the parent table calc separately from the child table calc, whereas a calculation containing two independent table calculation functions will act as a single table calc when determining how to compute the table calc,

In the Below Example, we are having two table calculations
1)Lookup   2)Index
While addressing we need to set adressing Options individually for both the functions.

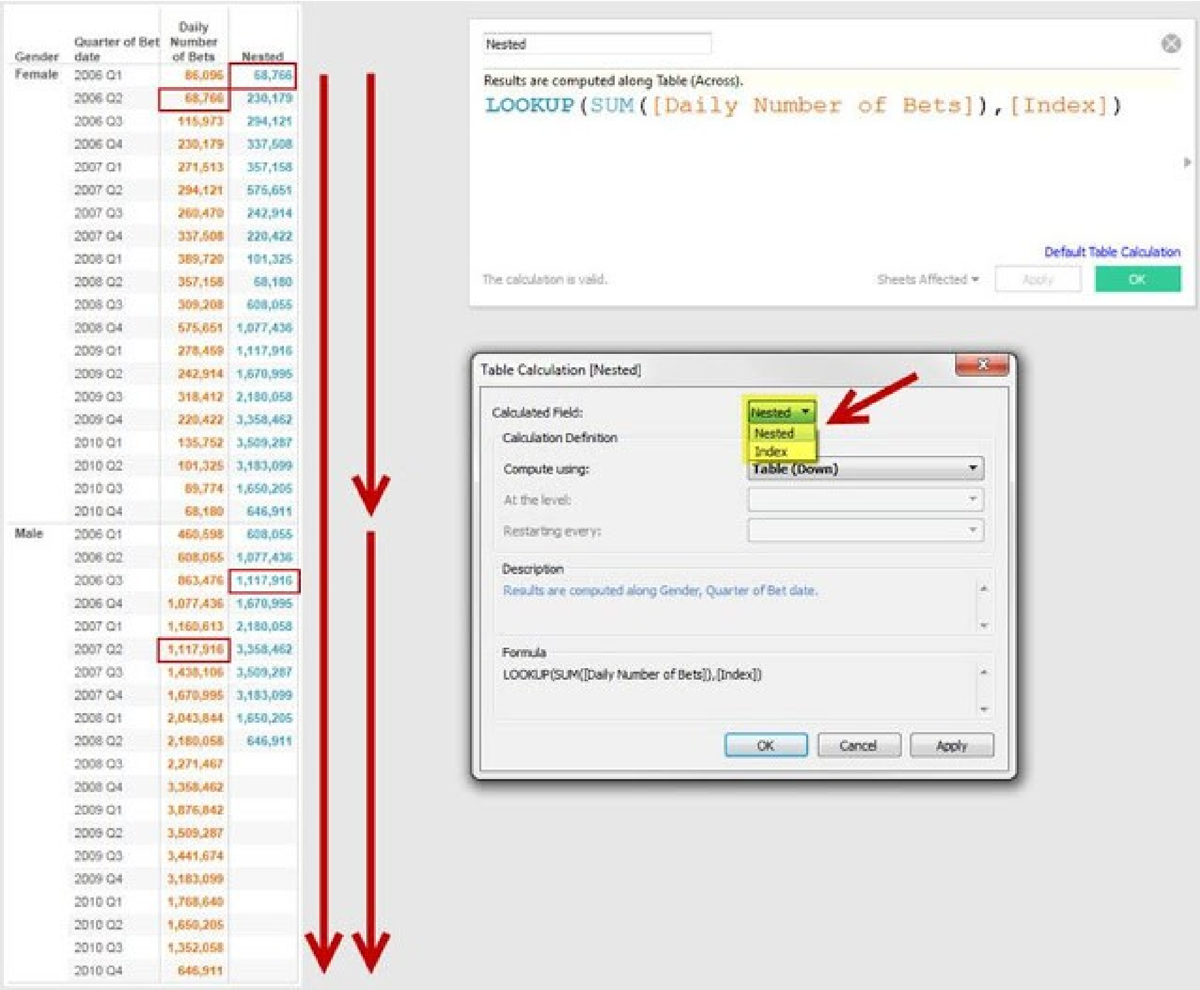
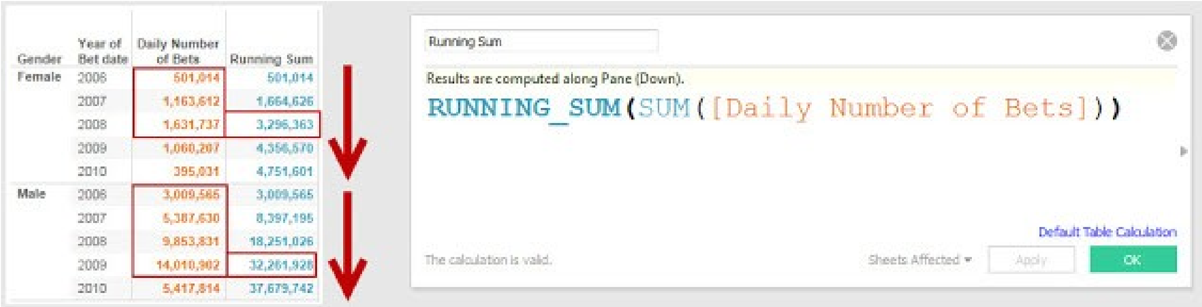
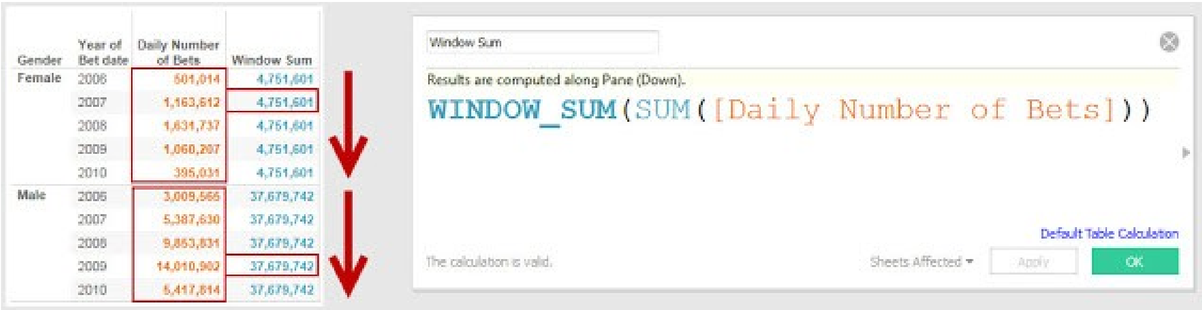


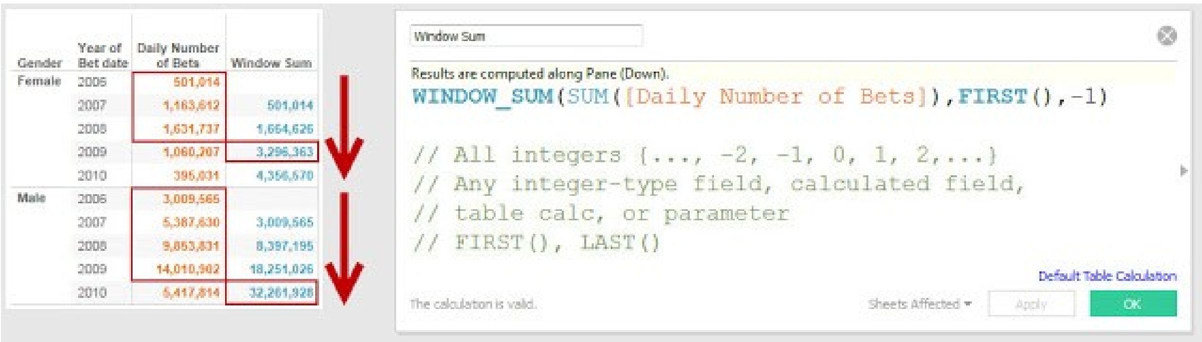
Table Calculation Functions

1.WINDOW,RUNNING FUNCTIONS:-

WINDOW functions perform aggregations on a selected range of data that is included in the structure of the view. The range is set by negative integers moving backward from a point and positive integers moving forward from a point. The arguments FIRST() and LAST() as well as parameters and other fields can also be used to include all data from a given point to the beginning or end of the data set.

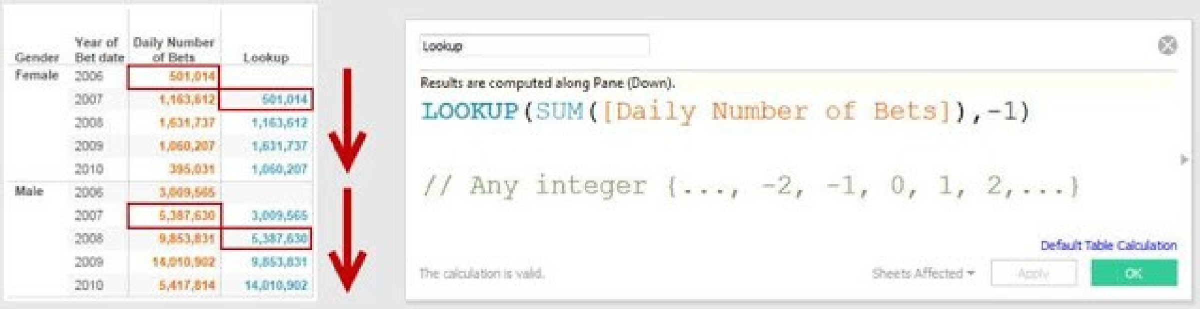


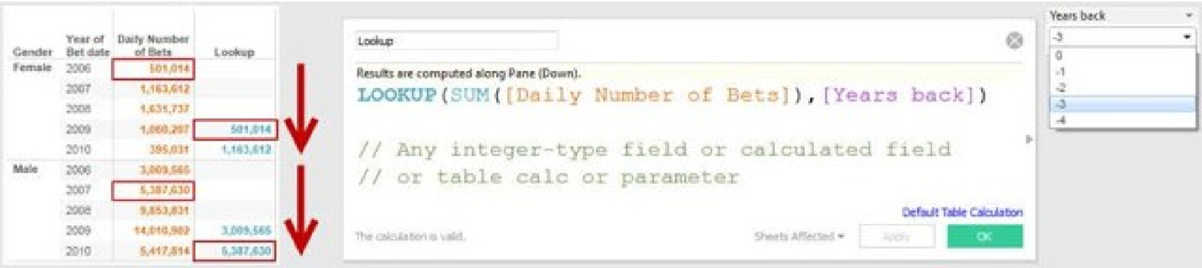




2.LOOKUP:-

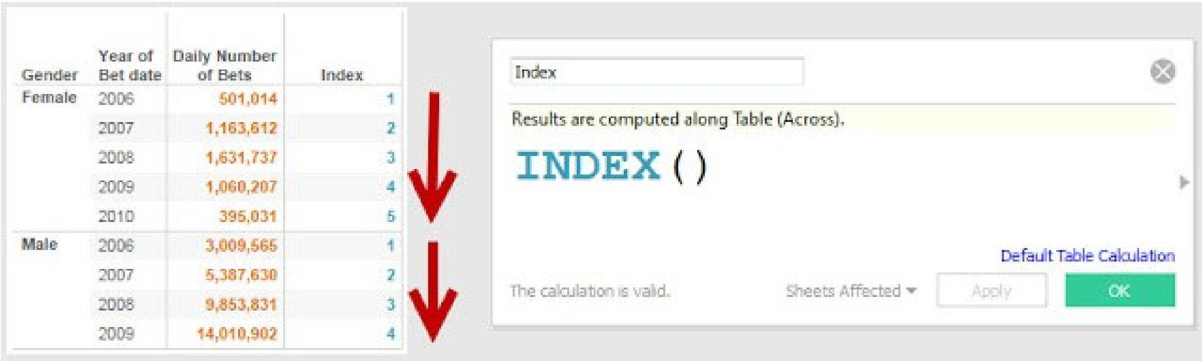
LOOKUP looks forward (positive integers) or backward (negative integers) a specified number of dimension values in the view. Backward and forward can be determined either by the structure of the view (i.e., left to right or top to bottom) or by a specified measure to sort



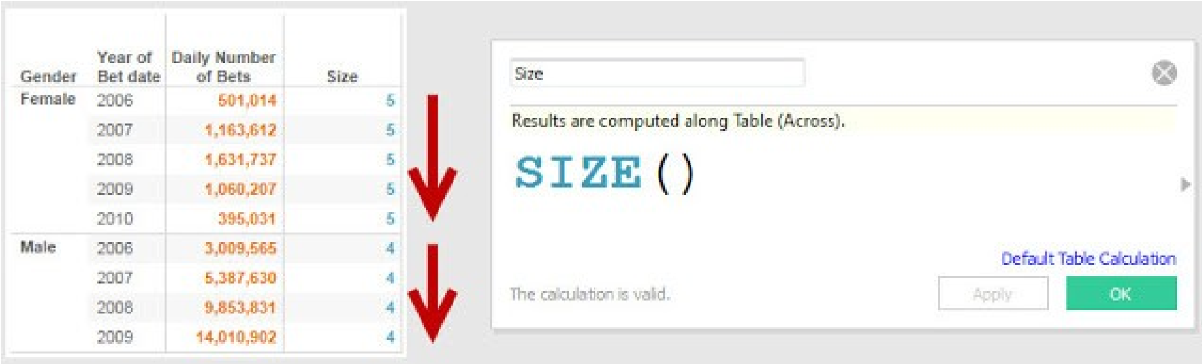


3.INDEX&SIZE:-

INDEX counts 1, 2, 3,… up through however many values there are in a given dimension or combination of dimensions.
SIZE tells you the number of values in a given dimension or combination of dimension. You can think of SIZE as the maximum INDEX value.



Gender	Year of Bet date	Daily Number of Bets	Index
Female	2006	501,014	1
	2007	1,163,612	2
	2008	1,631,737	3
	2009	1,060,207	4
	2010	395,031	5
Male	2006	3,009,565	1
	2007	5,387,630	2
	2008	9,853,831	3
	2009	14,010,902	4



Gender	Year of Bet date	Daily Number of Bets	Size
Female	2006	501,014	5
	2007	1,163,612	5
	2008	1,631,737	5
	2009	1,060,207	5
	2010	395,031	5
Male	2006	3,009,565	4
	2007	5,387,630	4
	2008	9,853,831	4
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